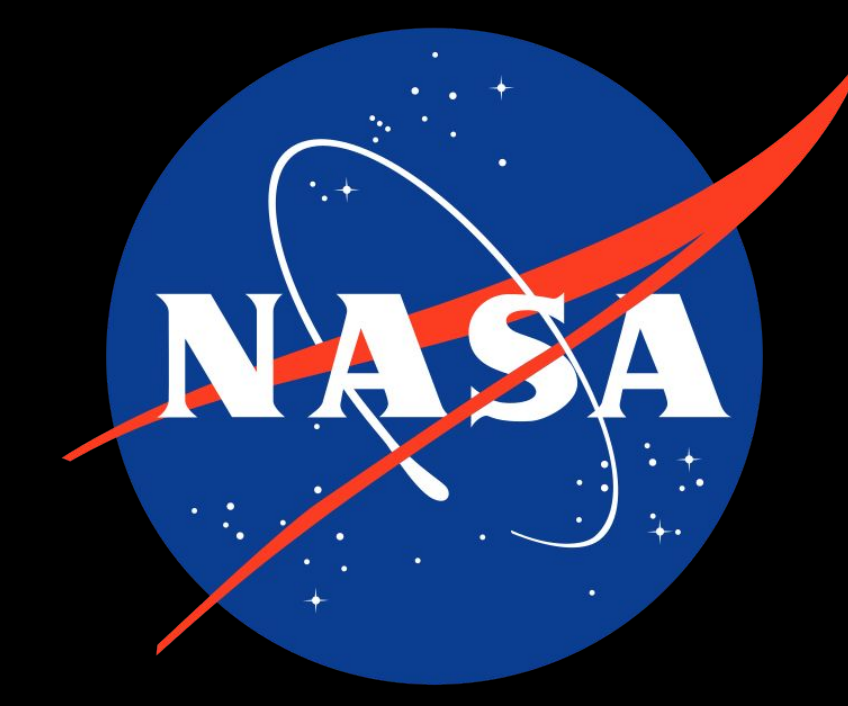


# Comparing Single-Scatter Neutron Spectrometers and the nDURE Instrument for Detecting High Energy Albedo Neutrons

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## Neutron Spectrometers

The scattering processes related to solar energetic particles (SEPs) and galactic cosmic rays (GCRs) striking the lunar and martian surface often produce neutrons as secondary particles. Able to possess energies greater than 100 MeV, these neutrons can pose a serious radiation risk to equipment and personnel. The ability to characterize the energy spectra of these albedo neutrons is becoming increasingly more valuable as NASA and other governmental and commercial space organizations prepare for a lunar base and further martian exploration. We compare the Neutron Dosimeter for Unknown Radiation Environments (nDURE) instrument, a highly precise double-scatter neutron spectrometer that is able to kinematically reconstruct incident neutron energies [Figure 1], with a similar single-scatter neutron spectrometer. While double-scatter neutron spectrometers are highly precise, they suffer from low effective neutron cross sections and higher complexity, whereas single-scatter neutron spectrometers are easier to build and offer much higher statistics, at the cost of poor performance at higher energies. We compare the energy resolution, energy orthogonality, and unfolding performance of both the nDURE instrument and a single-scatter neutron detector, finding that only the nDURE instrument gives meaningful unfolded spectra at higher energies.

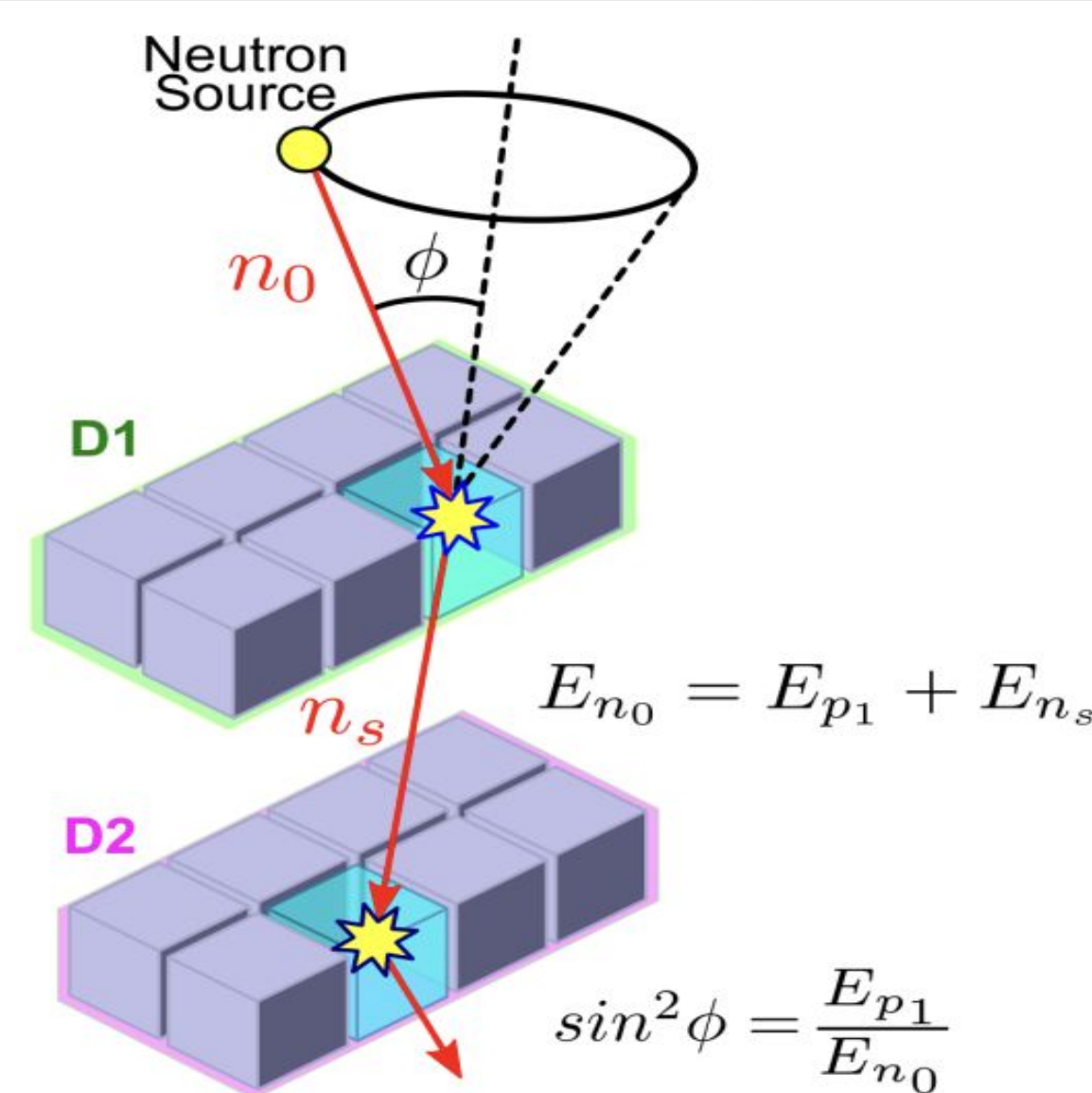


Figure 1: Energy reconstruction for a double-scatter spectrometer (LOONS) [1]

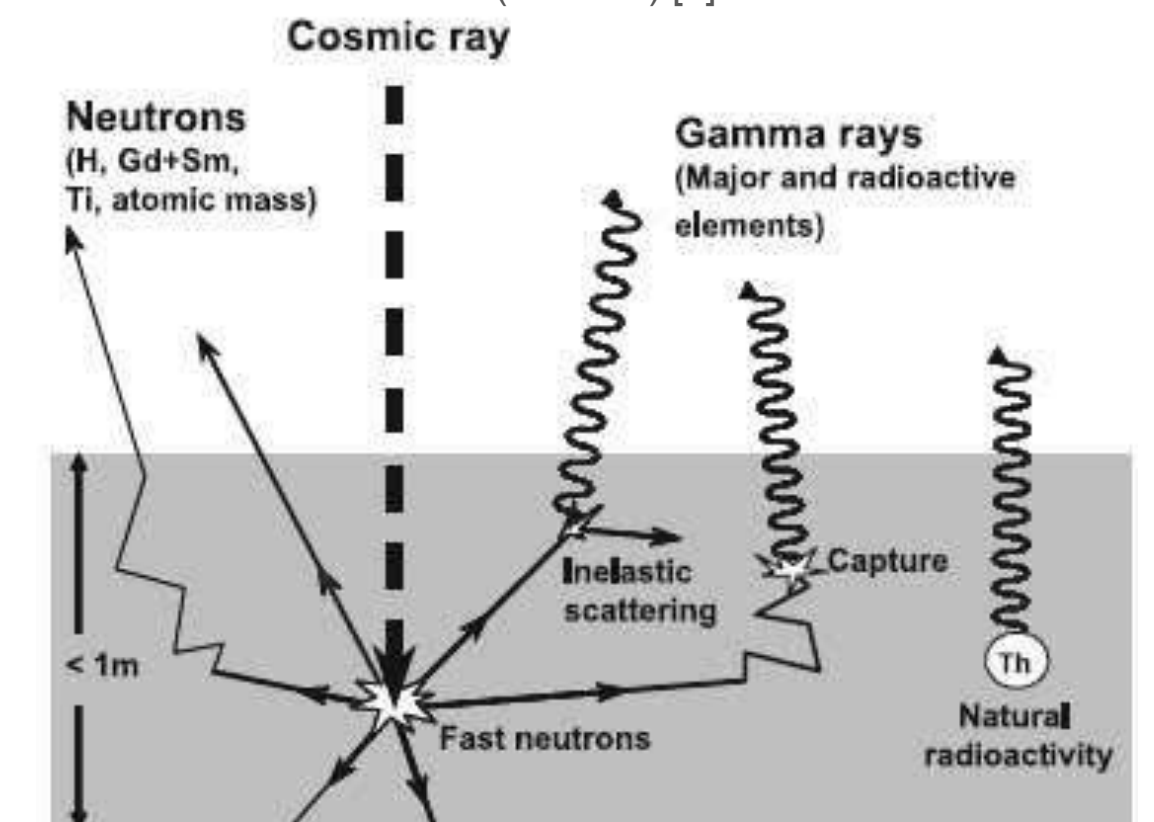


Figure 2: Galactic Cosmic Ray (GCR) scattering in the lunar regolith and releasing secondaries, including fast neutrons. [8]

## Simulation Pipeline and Energy Response Characterization

In order to compare the nDURE and single-scatter instruments, GEANT-4 was used to simulate each neutron detector [Figure 3]. These simulations allow us to characterize many detector features, including energy resolution [Figure 4]. We see as energy increases, the nDURE instrument's energy resolution decreases. This is due to the detector's time-of-flight (TOF) resolution having an increasingly worse effect as the neutrons travel faster. While it appears that the single-scatter instrument has excellent resolution, the energy curve is misleading, as in reality it is measuring "edge sharpness" and not energy resolution, since there is no distinct energy signal. Simulations reveal that the single scatter instrument suffers from poor peak separation and signal statistics [Figure 10], which make it difficult to tell different-energy neutrons apart in its reconstructed spectra.

Figure 3 (Below): Simulation flow chart. Demonstrates how the simulation pipeline takes as input a true distribution (left) and outputs a reconstructed distribution (right)

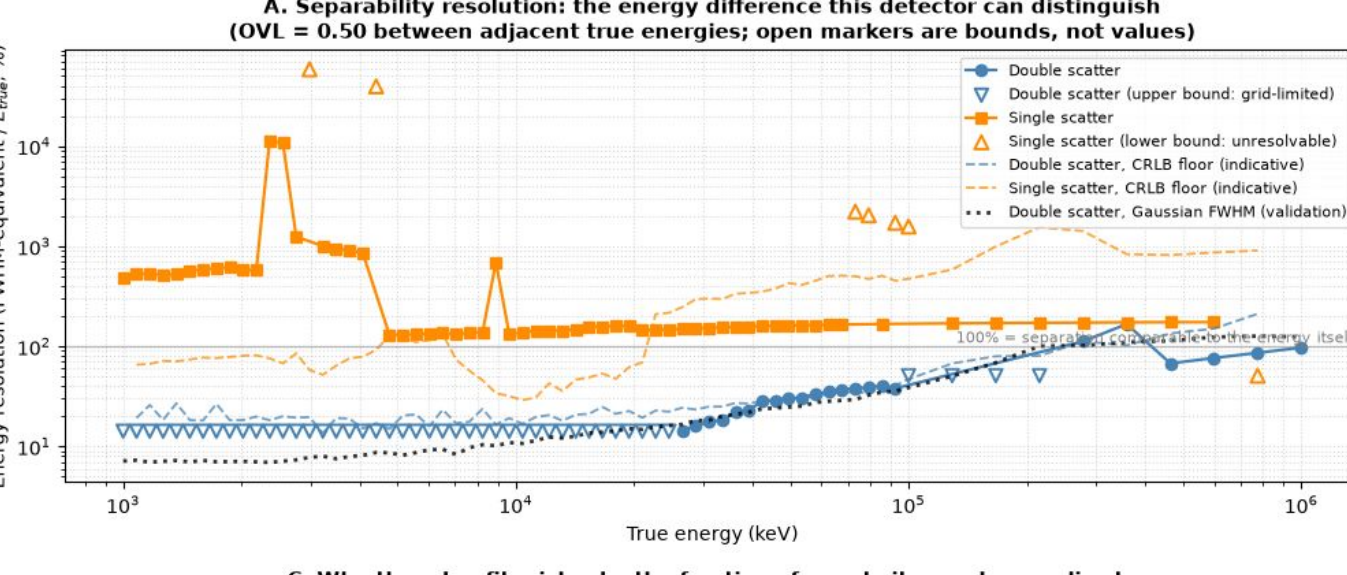
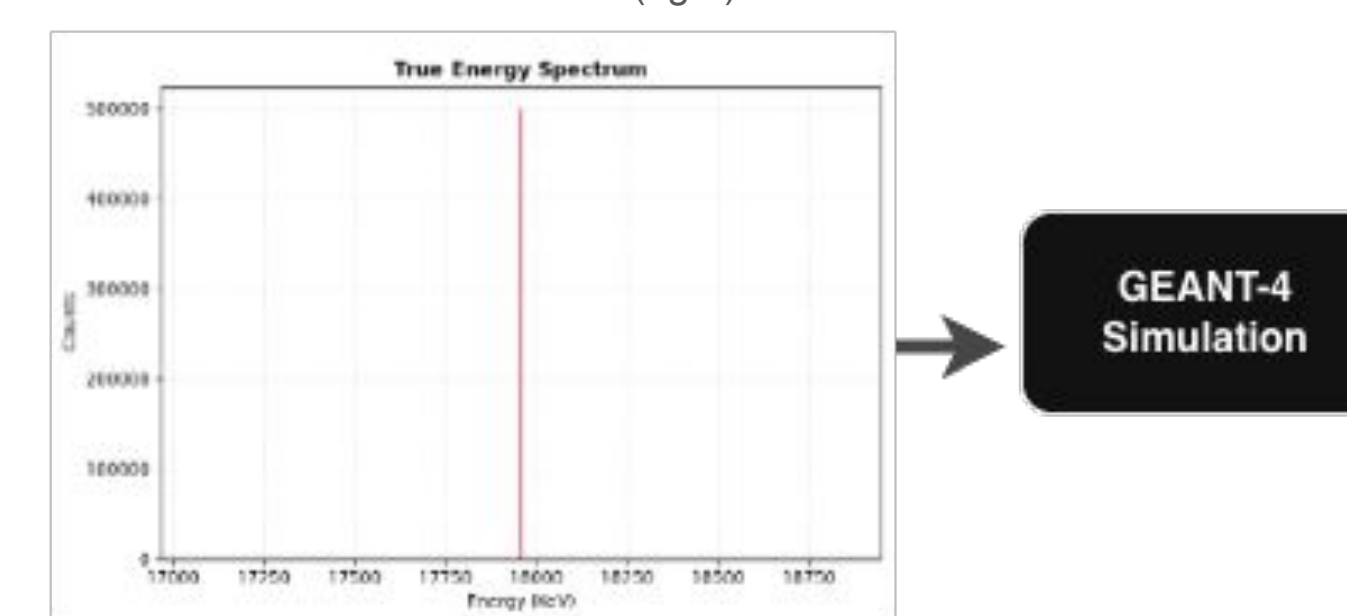


Figure 10: Topology-fair energy resolution. Panel A: Peak separability resolution. Panel B: Signal statistics as a function of true energy.

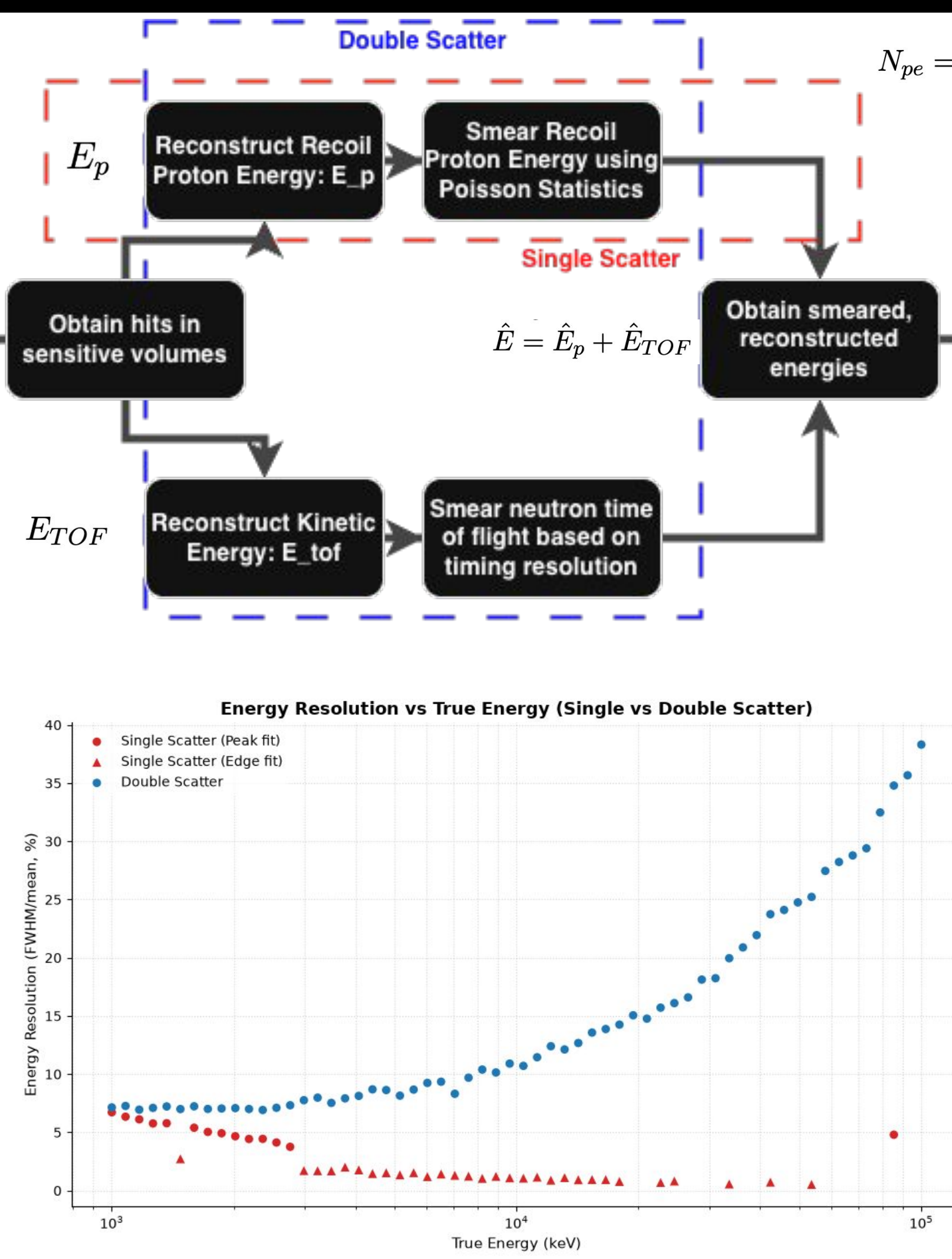


Figure 4: Energy resolution as a function of energy for the nDURE and single-scatter instruments.

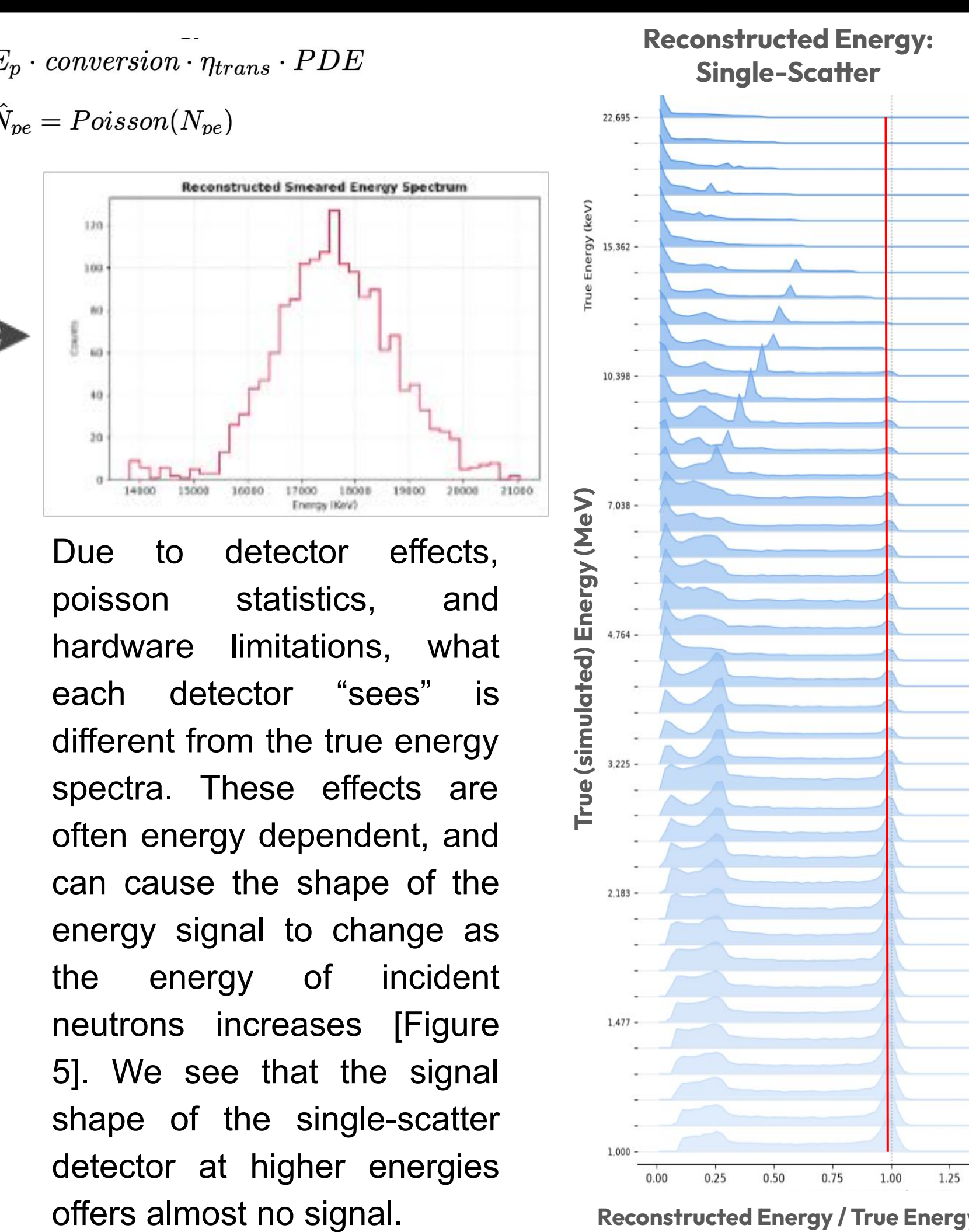


Figure 5: Energy ridgelines for the single-scatter instrument. As simulated energy increases, the distributions suffer from greater "smearing".

## A Simulated Comparison of Unfolding Quality: nDURE vs Single-Scatter

### Unfolding Premise

The energy spectra that we read from a neutron detector is not necessarily the true, physical energy spectra. Statistics, detector geometry, electronic noise, and hardware limitations can cause neutrons of a certain true energy to be measured or "reconstructed" at a different energy in a detector. Formally, we want to obtain the true distribution  $\mu$ , given a response matrix  $R$  and reconstructed distribution  $\nu$ . We can think of this as a vector equation, shown below:

$$\mu \times R = \nu$$

$$R\mu = \nu$$

$$\mu = R^{-1}\nu$$

By "inverting" the response matrix, we can make a statistical guess at the true distribution  $\mu$ . To invert these matrices, we use iterative bayesian unfolding (IBU), which makes use of bayes theorem to balance mathematical and physical accuracy via number of iterations:

$$P(\mu|\nu) \propto P(\nu|\mu) \cdot P(\mu)$$

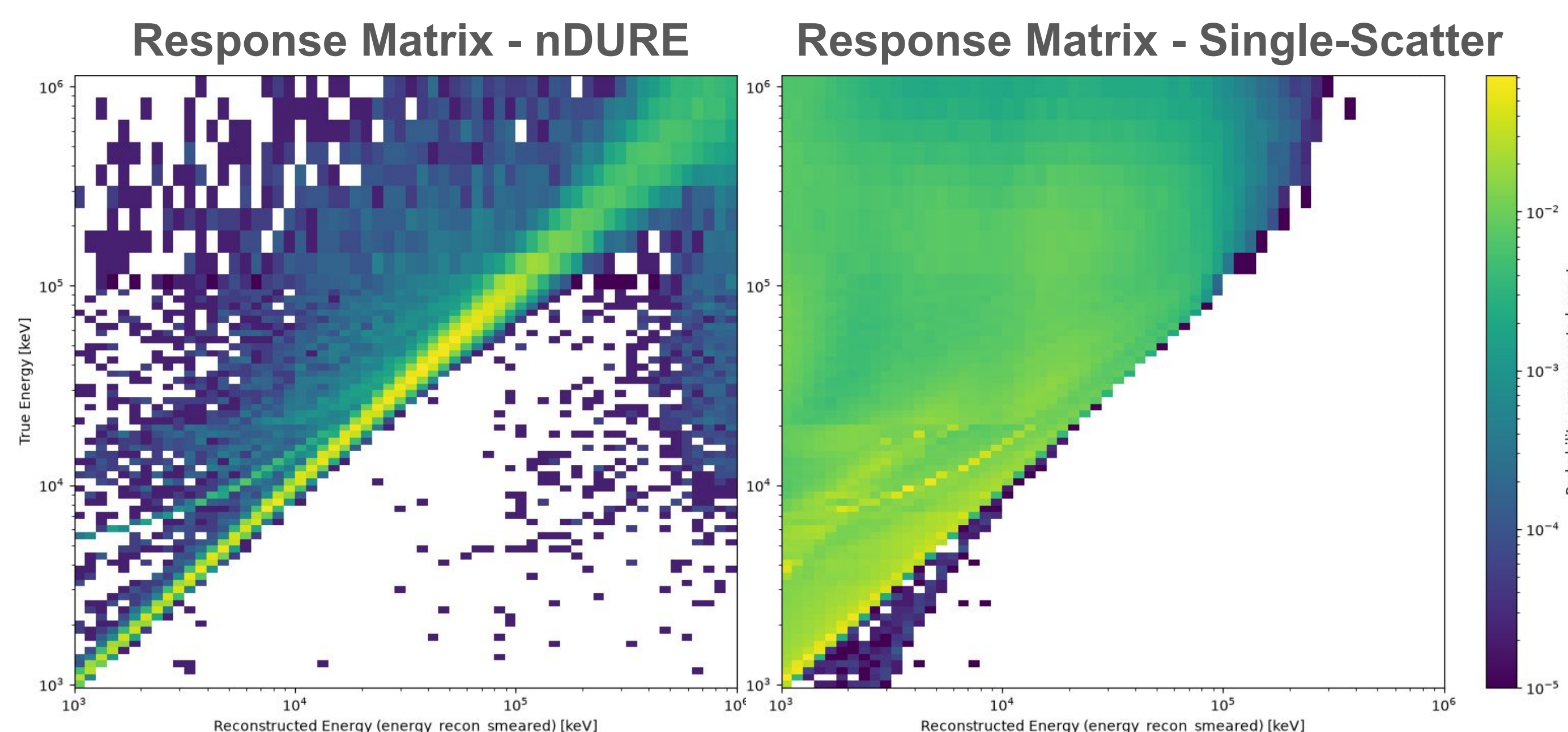


Figure 6: The monte-carlo simulated response matrices for nDURE and the single-scatter instrument. Each ranges from 1 MeV to 1 GeV. The x-axis represents reconstructed energies, and the y-axis represents true energies. Each cell value is in units of probability.

In figure 6, each response matrix was generated using monte carlo simulations of 70 different monoenergetic beams ranging from 1 MeV to 1 GeV passing through each detector. Each cell in the response matrix can be interpreted as the probability of a particle being reconstructed in bin  $i$ , given it has a true energy in bin  $j$ :

$$\nu_i = \sum_j P(i|j)\mu_j$$

$$\nu_i = R_{ij}\mu_j$$

Comparing the response matrices of nDURE and the single-scatter detector, we see they vary drastically in structure. We see the strong diagonal structure of nDURE's response matrix indicates a high probability of accurate reconstruction. The single-scatter's lack of diagonal structure suggests the opposite, where a 5 MeV and 10 MeV particle could be (incorrectly!) reconstructed in the same bin.

## Unfolding Results and Future Investigation

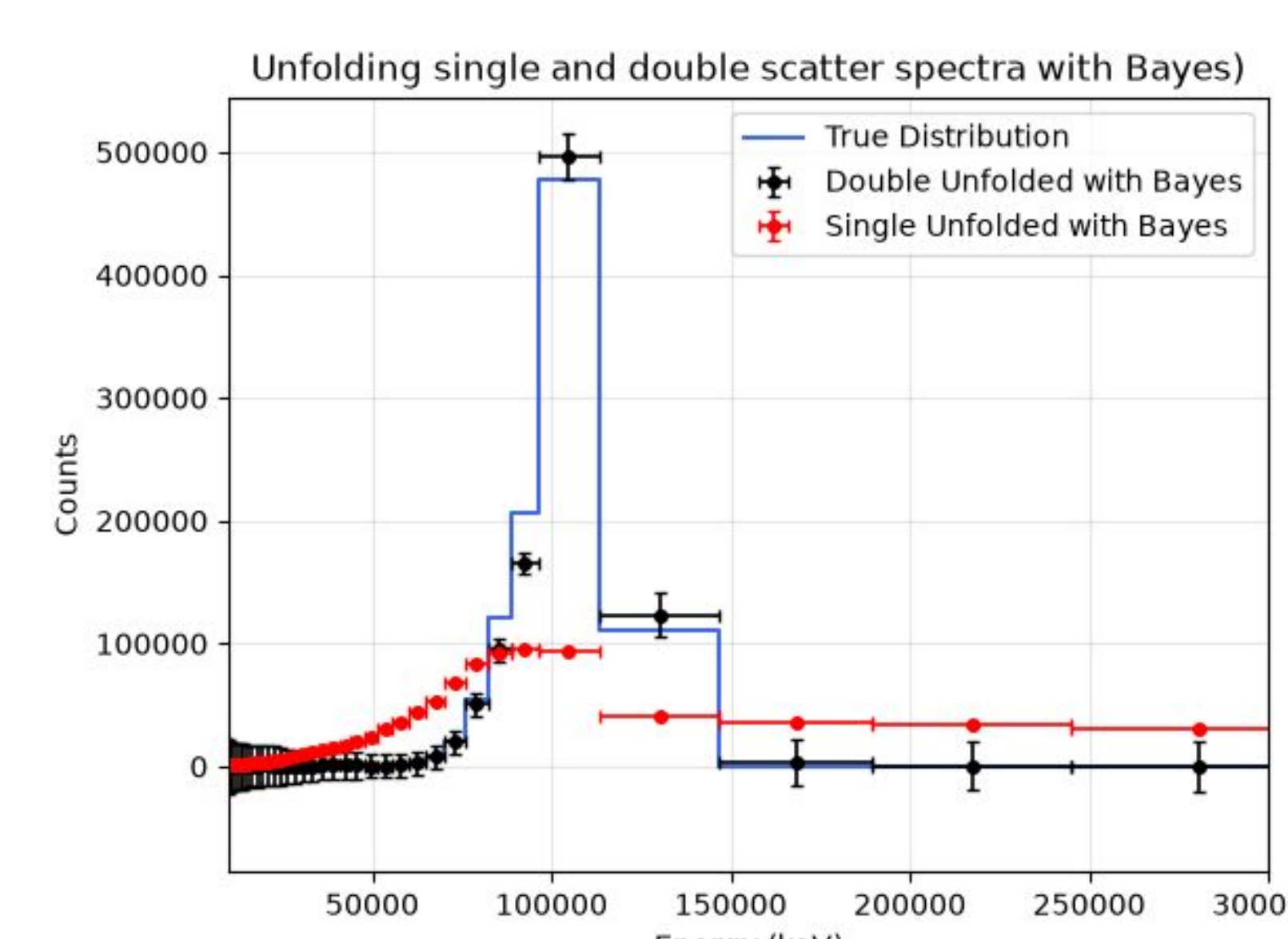


Figure 7: Histogram of a simulated, true gaussian distribution centered at 100 MeV, with the same distribution unfolded using spectra from nDURE and the single-scatter instrument. We see nDURE unfolds closer to the original spectra.

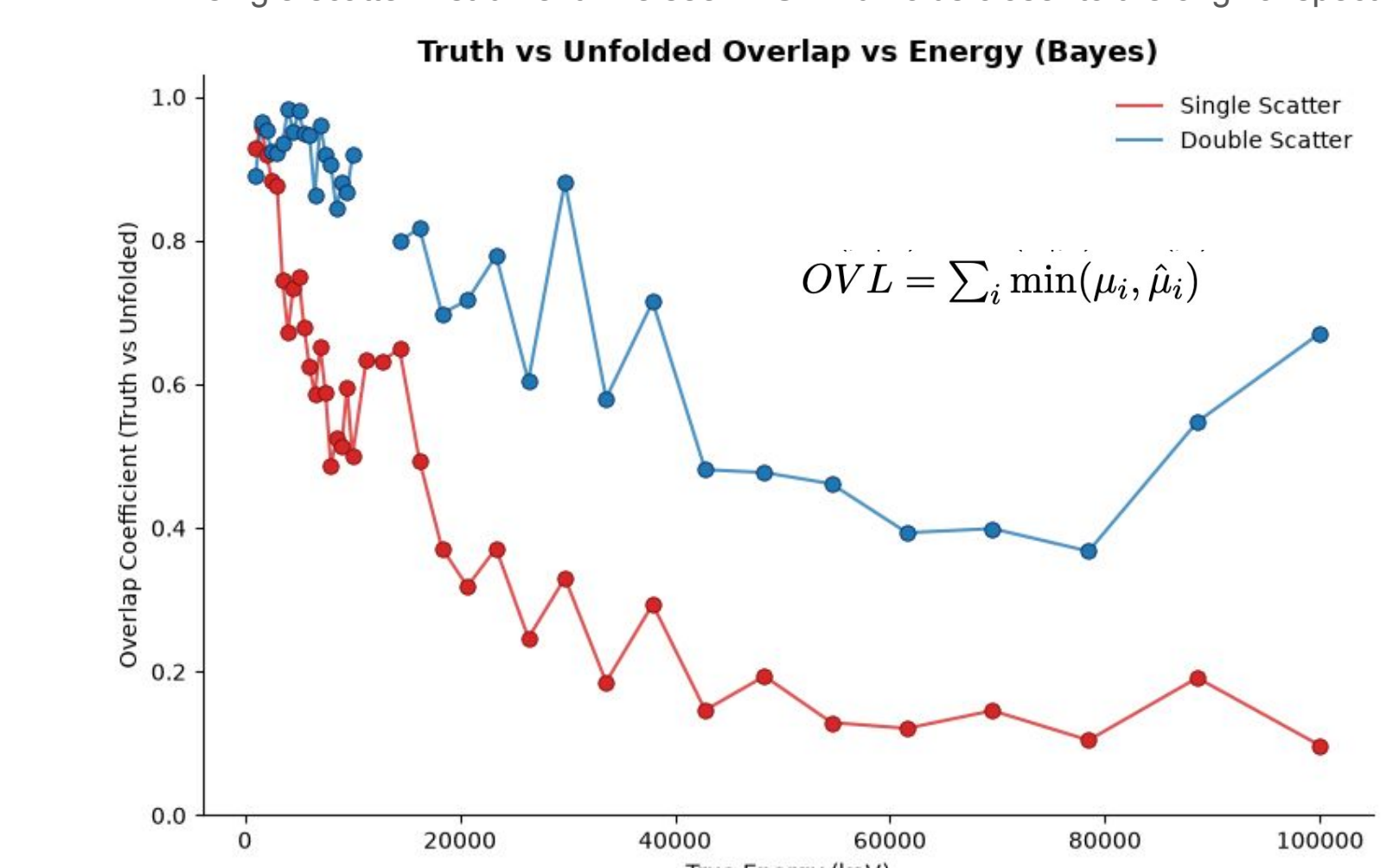


Figure 8: Truth vs unfolded spectra overlap as a function of true energy. Each point represents the percent overlap of a gaussian and its unfolded counterpart for each instrument.

A comparison of the unfolding power of each instrument's reconstructed energy spectra was made in-simulation by passing beams of neutrons with gaussian energy distributions through each instrument, ranging from 1-100 MeV. Once reconstructed energy spectra were obtained for each gaussian, they were unfolded using iterative bayesian unfolding. An example of the unfolded spectra for a gaussian with mean energy 100 MeV can be seen in Figure 7. Using these simulations, we observe that both instruments have a comparable unfolding capability (accurately recovering the true gaussian) at low energies (< 5 MeV). However, as the mean energy of the simulated neutron beam grows past 5 MeV, we see the performance of the single-scatter instrument begins to degrade dramatically, with reconstructed distribution percent overlap staying below 60% and approaching 0% at higher energies [Figure 8]. This suggests that a double-scatter instrument such as nDURE is necessary for characterizing unknown albedo neutron spectra, as the results of a single-scatter instrument past 5 MeV are not well-constrained.

### Future Investigation

It is important to note that all simulation results we observe are instrument-specific, and other double-scatter or single-scatter instruments may have different performance characteristics. Future research will involve the use of a laboratory neutron source to obtain experimental data, and the assessment of another technique to obtain true spectra known as forward-folding [Figure 9].

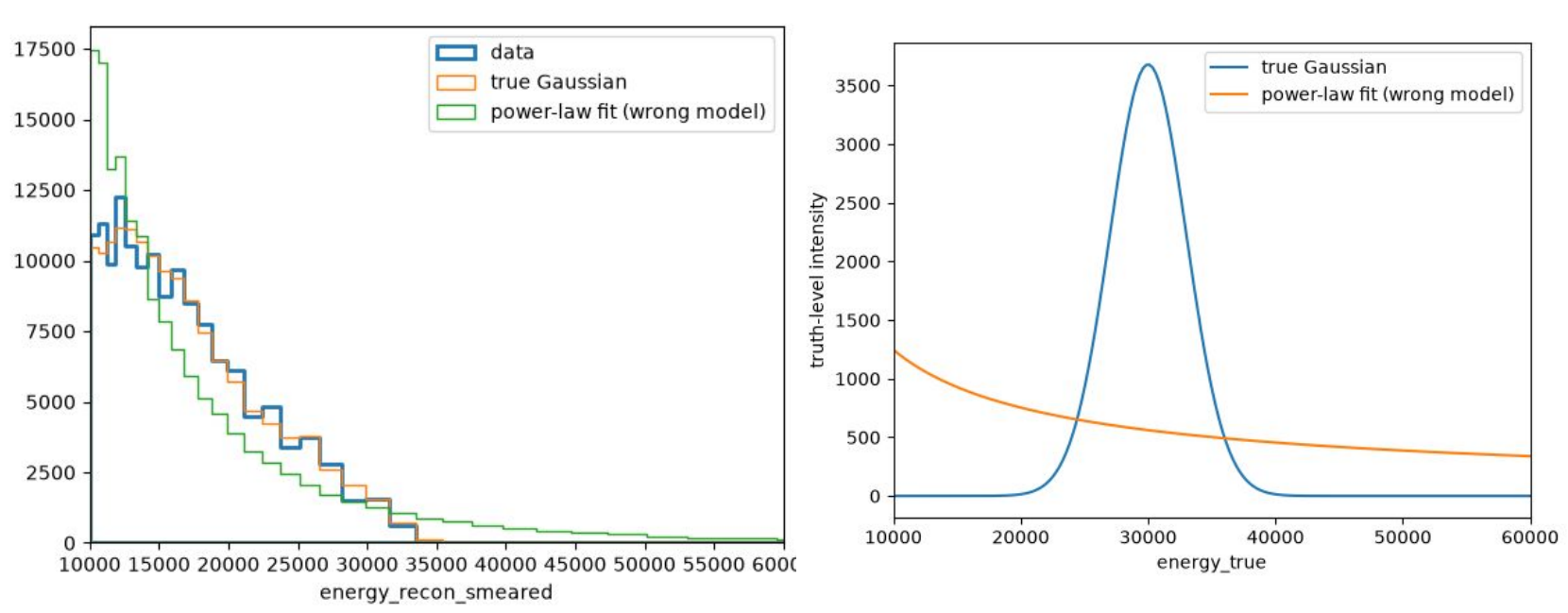


Figure 9: An example of forward-folding being used to obtain true energy spectra. Forward-folding can be a powerful tool when a general idea of the underlying neutron energy spectra is known.

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### References

